

MarketCatalyst — Single-Vendor Consolidation Analysis

Question: If we wanted **one** market-data vendor to supply **100%** of what the MarketCatalyst app needs, what is missing / what would we have to ask that vendor to provide?

Two scenarios are analysed:

- **Scenario A — Polygon only**
- **Scenario B — FMP only**

This document is grounded in a full code inventory of the running app (every Polygon, FMP, FRED and SEC-EDGAR call), **not** on a generic vendor comparison. Date: 2026-08-18.

0. Bottom line (read this first)

	Polygon-only	FMP-only
Can it reach 100% as the app is designed today?	No — even with every paid upgrade	Almost — ~95%+, one true blocker
Hard blockers (no product exists at the vendor)	Consensus/ non-GAAP EPS & estimates , analyst ratings & price targets, institutional/13F rollup, earnings-call transcripts, forward earnings calendar	Short interest only
Solvable by paying/upgrading the same vendor	Real-time (drop 15-min delay), real index & VIX levels, short interest, crypto, >5yr history, non-experimental financials	Real-time tick stream, deeper intraday history, options depth
Net verdict	Not viable without dropping ~4 feature areas (Analyst Actions, Earnings estimates, Institutional, Transcripts). Polygon does not sell that data.	Viable. FMP is effectively a super-set of what we use; the work is <i>wiring</i> the price/quote/macro/commodity feeds we currently take from Polygon/FRED/SEC. Short interest stays unsolved (it's unsolved today too).

One-line summary: Polygon is a market-**infrastructure** vendor (prices, bars, corporate actions) and has no analyst/estimates/ownership products — so Polygon-only cannot power the fundamentals side of the app. FMP is a **fundamentals + analyst + everything** vendor that also carries prices, indices, commodities and macro — so FMP-only is achievable except for short interest.

1. Current architecture — who supplies what today

The app uses **exactly four** external sources. Everything else is computed in our own jobs.

Vendor	Plan / access	Supplies today
Polygon.io	Stocks Starter , 15-min delayed, 5-yr rolling history	Prices/quotes/bars/snapshots, Top Movers (grouped daily), dividends & splits, related-companies (peers), market status & holidays, treasury yields (/fed/v1), GAAP financials (/vX/reference/financials), IPO calendar, options contracts, news (+sentiment insights), ticker details (name/sector-via-SIC/market-cap/description/logo), full ticker universe & search
FMP	/stable/ API (v3/v4 deprecated → 403)	Non-GAAP earnings (actual-vs-estimate, the beat/miss basis), forward EPS/revenue estimates, upcoming earnings calendar, analyst consensus / grades / price targets, institutional (13F) ownership rollup , earnings-call transcripts, forward economic calendar, cap-weighted sector performance, stock news
FRED	free key	Header tape: WTI (DCOILWTICO), Bitcoin (CBBTCUSD); 12 macro series (CPI, Core CPI, unemployment, payrolls, fed funds, PPI, retail sales, jobless claims, GDP, industrial production, consumer sentiment, DGS10); macro-regime inputs (yield-curve spread, VIX, HY credit spread, S&P level, unemployment)
SEC-EDGAR	free (public)	Form 4 insider transactions , 13F fund holdings (fund-indexed, CUSIP-level), 8-K filings wire + earnings announcements (item 2.02 → BMO/AMC + reaction), S-1/424B IPO registration pipeline

Everything computed in-house (from Polygon bars): RS rating, technical rating, RSI/MACD/SMA/EMA/Stochastic/ADX, beta, 52-week hi/lo, realized vol, key-level pivots, relative volume, Fear & Greed, market breadth/internals (advancers, decliners, TRIN, McClellan), recaps, sector ranks.

Gaps that already exist today (no vendor supplies them)

These are missing **regardless** of consolidation — the honest baseline:

- **Short interest** — no field anywhere (Polygon 404s, FMP has no product).
- **Guidance / implied move / post-earnings reaction** — fields exist on the earnings calendar but are always null.
- **Revenue (sales) surprise** — FMP [/stable/earnings](#) is EPS-only, so the Sales %surp column renders "—".
- **Real index levels** (SPX/NDX/DJI/RUT), **spot gold**, **spot VIX**, **DX** — approximated with ETF-proxy × multiplier because Polygon index snapshots are [NOT_AUTHORIZED](#) on Starter (gold specifically = GLD × 10.89).

2. Scenario A — Polygon only

2.1 What Polygon already covers (no change needed)

Prices/quotes/bars, movers, dividends, splits, peers, market status/holidays, treasury yields, GAAP financials, IPO calendar (priced), options, news, company profile/sector/logo, ticker universe/search, and everything we compute from bars (technicals, RS, pivots, breadth, Fear & Greed). This is the majority of the *screens*, but **not** the fundamentals depth.

2.2 Solvable by paying / upgrading Polygon (product exists — just entitle it)

Ask Polygon to enable these; they exist in Polygon's catalog:

Need	Polygon product to request	Removes today's hack
Real-time (drop 15-min delay)	Real-time Stocks plan + real-time WebSocket (wss://socket.polygon.io)	Currently NOT_AUTHORIZED ; uses delayed.polygon.io
Real index levels + VIX	Indices cluster / add-on (I:SPX , I:NDX , I:DJI , I:RUT , I:VIX)	Removes SPY×10 / QQQ×36.3 / DIA×100 / IWM×10 ETF proxies and gives a real VIX

Need	Polygon product to request	Removes today's hack
Short interest	Short Interest / Short Volume endpoint (add-on)	Fills the one field that's blank today — <i>Polygon can do this, FMP cannot</i>
Bitcoin / crypto	Crypto cluster (<code>X:BTCUSD</code>)	Removes FRED <code>CBBTCUSD</code> dependency
History > 5 years	Higher history tier	Removes <code>planHistoryFloor()</code> clamp
Stable financials	<code>/stocks/financials/v1/*</code> — needs Advanced or Financials add-on	Off the experimental <code>/vX/</code> namespace (still GAAP, see blocker below)

2.3 Hard blockers — Polygon has no product for these

No amount of upgrading fixes these; Polygon simply does not sell this data. Each maps to a live feature:

Missing data	Feature that breaks	Why Polygon can't
Consensus / non-GAAP EPS actual (the app's whole EPS/P/E/growth/beat-miss basis, matching NASDAQ/IBD)	EPS everywhere, P/E, EPS growth, beat/miss, Earnings Playbook	Polygon financials are GAAP from SEC filings only. It has no consensus/adjusted-EPS product. Using Polygon GAAP produces the spurious huge beats/misses we specifically moved off (PANW case).
Forward EPS/revenue estimates ("what street expects", <code>annualEstimates</code> , upcoming-calendar estimates)	Earnings Hub estimates, forward <code>*YYYY</code> columns	Polygon has no estimates/consensus feed at all.
Analyst ratings — consensus buy/hold/sell, grades, upgrades/downgrades	Analyst Actions screen, consensus card	Polygon has no analyst product.
Price targets — consensus high/low/avg/median, per-firm targets, PT trend	Price-target cards & upside	Polygon has no price-target product.
Institutional / 13F ownership rollup (ticker-indexed % owned, holder counts)	Institutional card & screen	Polygon has no ownership/13F product.
Earnings-call transcripts	Transcript viewer / "Earnings call"	Polygon has no transcripts.
Forward earnings calendar (upcoming report dates for the whole market)	Earnings Hub calendar (upcoming)	Polygon only knows <i>past</i> report dates (via <code>filing_date</code>). No forward calendar.
Forward economic calendar w/ consensus estimates + broad macro series	Macro screen "this/next week" + most macro tiles	Polygon <code>/fed/v1</code> has only treasury yields & inflation — not payrolls/GDP/PPI/retail/claims/sentiment, and no forward release schedule with estimates.
Commodities spot (WTI, gold)	WTI & Gold tape tiles	Polygon has no commodities/futures product; only ETF proxies (USO/GLD).

Still-free fallbacks that are NOT Polygon (only relevant if "Polygon only" is taken literally): insider Form 4, 13F fund positions, 8-K wire, IPO registration pipeline all come from **SEC-EDGAR** (free/public). Broad macro comes from **FRED** (free). If "Polygon only" means "drop FRED & SEC too," add insider, 13F drill-down, filings wire, IPO pipeline and macro to the blocker list. SEC/FRED can *partially* substitute (insider, 13F, GAAP financials, macro history) but **cannot** supply estimates, analyst ratings, transcripts, or non-GAAP EPS.

2.4 Polygon "ask list" (what to request from Polygon)

- 1. Real-time entitlement (stocks) + real-time WebSocket.
- 2. **Indices** add-on (SPX/NDX/DJI/RUT + **VIX**).
- 3. **Short Interest / Short Volume** endpoint.
- 4. Crypto cluster (BTC).
- 5. History beyond 5 years.
- 6. `/stocks/financials/v1/*` (Advanced/Financials add-on).
- 7. **Ask whether Polygon has any roadmap for: analyst estimates/ratings, price targets, institutional/13F, transcripts, forward earnings & economic calendars, commodities.** (*Expected answer: no — these are outside Polygon's product line.*)

2.5 Verdict — Polygon only

Not viable for the app as designed. Polygon can be made excellent for *price/market-structure* data (and can uniquely add short interest and real index/VIX levels), but it fundamentally cannot power **Analyst Actions, forward Earnings estimates, Institutional ownership, or Transcripts**, and cannot supply the **non-GAAP EPS** basis the whole fundamentals layer is built on. Going Polygon-only means deleting those features or bolting on SEC/FRED (which still can't cover estimates, ratings, or transcripts).

3. Scenario B — FMP only

FMP is a full-stack financial-data API: it already gives us the hardest parts (estimates, analyst, ownership, transcripts). To go FMP-only we mostly need to **move the price/market/macro feeds we currently take from Polygon/FRED/SEC onto FMP endpoints that already exist.**

3.1 What FMP already covers (live in the app today)

Non-GAAP earnings & surprises, forward estimates, upcoming earnings calendar, analyst consensus/grades/price-targets, institutional 13F rollup, transcripts, forward economic calendar, sector performance, news. This is the entire fundamentals + analyst + ownership layer — the part Polygon can't do.

3.2 What must be *wired* onto FMP (FMP has the product; app just doesn't call it yet)

These are currently Polygon/FRED/SEC. FMP has equivalents — this is integration work, not a vendor gap:

Need (currently)	FMP endpoint(s) to add	Notes / upside
Quotes & prices (Polygon snapshot)	<code>/stable/quote</code> , <code>/stable/batch-quote</code>	Delayed/EOD on lower tiers; real-time needs a higher plan
Daily & intraday bars (Polygon aggs)	<code>/stable/historical-price-eod/*</code> , <code>/stable/historical-chart/{interval}</code>	Intraday history depth is plan-limited
Real index levels + VIX (proxies today)	<code>/stable/quote?symbol=^GSPC,^IXIC,^DJI,^RUT,^VIX</code>	Solves the ETF-proxy hack for free — real S&P/Nasdaq/Dow/Russell/VIX
Commodities spot WTI, gold (FRED/GLD proxy)	<code>/stable/quote</code> for commodity symbols (WTI, gold)	Real spot — removes the GLD × 10.89 workaround
Bitcoin/crypto (FRED)	<code>/stable/quote</code> crypto (<code>BTCUSD</code>)	Removes FRED <code>CBBTCUSD</code>
Treasury yields (Polygon <code>/fed/v1</code>)	<code>/stable/treasury-rates</code>	—

Need (currently)	FMP endpoint(s) to add	Notes / upside
Macro series (FRED, 12 series)	/stable/economic-indicators (GDP, CPI, unemployment, etc.)	Replaces FRED observations; verify each series is offered
Movers (Polygon grouped daily)	/stable/biggest-gainers , /stable/biggest-losers , /stable/most-actives	—
Market hours / holidays (Polygon)	/stable/exchange-market-hours , holidays	—
Dividends & splits (Polygon)	/stable/dividends , /stable/splits	—
Company profile/sector/logo (Polygon)	/stable/profile (has description, sector, industry, beta, image)	—
Peers (Polygon related-companies)	/stable/stock-peers	Curated list vs Polygon's algorithmic — quality differs
Insider Form 4 (SEC)	/stable/insider-trading/*	Replaces SEC Form 4 parsing
IPO calendar + pipeline (Polygon + SEC)	/stable/ipos-calendar , IPO prospectus/disclosure feeds	Verify pipeline (S-1/424B) coverage vs SEC
Ticker universe & search (Polygon)	/stable/stock-list , /stable/search-symbol	—
GAAP statements (Polygon /vX/)	/stable/income-statement , /stable/balance-sheet-statement , /stable/cash-flow-statement	FMP already gives us the non-GAAP EPS too
Technicals (computed from Polygon bars)	keep computing from FMP bars, or /stable/technical-indicators/*	No change to logic if we just swap the bar source
Breadth / Fear & Greed / recaps (computed)	compute from FMP EOD/index series	Heavier (needs full-market EOD) but feasible

3.3 Hard gaps / caveats with FMP-only

Item	Severity	Detail
Short interest	Hard blocker	FMP has no short-interest product on /stable . Unsolved today too. Would still need FINRA (or Polygon's short-interest add-on, ironically).
True real-time tick / NBBO	Medium	FMP is not a tick-data/market-infrastructure provider. Real-time quotes need FMP's paid WebSocket; there is no full trade/quote tick feed like Polygon's. Fine for the current 15-min-delayed design; a limitation if we ever want true real-time depth.
Options depth	Low	The app's options chain (8 curated tickers) comes from Polygon contracts. FMP options coverage is thin — may not fully replace it.
8-K filings wire + item-2.02 earnings-announcement classification & reaction	Medium	Uses SEC acceptance-time → BMO/AMC logic. FMP has SEC-filings/press-release RSS but not the same item-level 2.02 classification; would need re-implementation or approximation.
Per-firm price targets	Quality	Already flaky on FMP (price-target-news drops ~30% on name mismatch, errors swallowed to []). Not a missing product — a completeness issue.

Item	Severity	Detail
News sentiment	Quality	FMP article sentiment is frequently null (vs Polygon's per-ticker insights).
Silent empty 200s	Reliability	FMP returns empty 200 (not 429) under burst; we already pace requests and preserve prior values. Consolidating <i>more</i> load onto FMP raises the importance of that pacing/caching.
News redistribution licensing	Commercial	Already accepted for FMP news; confirm it covers broader redistribution if FMP becomes the sole feed.

3.4 FMP "ask list" (what to request / confirm with FMP)

1. **Short interest** data — do they have any product or roadmap? (*Likely no.*)
2. Plan tier that includes: **real-time (or acceptable delayed) quotes + WebSocket**, adequate **intraday history depth**, and **full-market EOD** (for breadth/movers/recaps).
3. Confirm coverage & symbols for: **index quotes** (^GSPC/^IXIC/^DJI/^RUT/^VIX), **commodities spot** (WTI, gold), **crypto (BTC)**, **treasury rates**, and each **economic indicator** we show (CPI, Core CPI, unemployment, payrolls, fed funds, PPI, retail sales, jobless claims, GDP, industrial production, consumer sentiment).
4. **IPO registration pipeline** (S-1/424B) coverage vs our SEC pipeline.
5. **Options** coverage for our 8-ticker chain (or accept dropping it).
6. **8-K / filings feed** granularity (item 2.02) — or accept computing earnings-announcement sessions/reactions ourselves from price + calendar.
7. Rate limits / burst behaviour for a single-vendor load; redistribution licensing for sole-feed use.

3.5 Verdict — FMP only

Viable (~95%+). FMP already owns the hard half (estimates, analyst, ownership, transcripts, earnings/econ calendars). The remaining work is integration: repoint prices, indices, commodities, crypto, treasury, macro, insider, movers, market-hours, IPO and universe onto FMP endpoints that already exist — and this actually **removes** several of today's hacks (real index & VIX levels, real spot gold, real BTC). The single unavoidable hole is **short interest**, which is a gap **today** anyway. Real-time tick data and options depth are the quality trade-offs to accept.

4. Side-by-side gap matrix

Legend:  native / already used ·  needs wiring or paid upgrade (product exists) ·  hard gap (no product)

Data area	Polygon-only	FMP-only
Real-time/delayed quotes & bars	 ( real-time = upgrade)	 wire ( real-time = upgrade)
Movers / grouped market		 wire (biggest-gainers ...)
Dividends & splits		 wire
Company profile / sector / logo		 wire (profile)
Peers / related companies		 wire (stock-peers)
Technicals / RS / pivots (computed)		 recompute from FMP bars
Index levels + VIX	 Indices add-on	 wire (^GSPC...^VIX) — real levels
Commodities spot (WTI, gold)	 (proxy only)	 wire (real spot)

Data area	Polygon-only	FMP-only
Bitcoin / crypto	🟡 crypto cluster	🟡 wire
Treasury yields	✅ (/fed/v1)	🟡 wire (treasury-rates)
Broad macro series	❌ (only treasury/inflation)	🟡 wire (economic-indicators)
Forward economic calendar + estimates	❌	✅
News (+sentiment)	✅ (sentiment ✅)	✅ (sentiment 🟡 often null)
GAAP financial statements	✅ (🟡 stable API = add-on)	🟡 wire
Non-GAAP / consensus EPS	❌	✅
Forward EPS/revenue estimates	❌	✅
Earnings surprises (beat/miss)	❌ (GAAP only → wrong)	✅
Forward earnings calendar	❌	✅
Analyst ratings / grades	❌	✅
Price targets (consensus + per-firm)	❌	✅ (per-firm 🟡 ~30% gaps)
Earnings-call transcripts	❌	✅
Institutional / 13F rollup	❌	✅
Insider (Form 4)	❌ (SEC today)	🟡 wire (insider-trading)
13F fund positions (CUSIP drill-down)	❌ (SEC today)	🟡 verify FMP fund-level 13F
8-K filings wire / earnings announcements	❌ (SEC today)	🟡 approximate / recompute
IPO calendar (priced)	✅	🟡 wire
IPO registration pipeline (S-1/424B)	❌ (SEC today)	🟡 verify FMP coverage
Market status / holidays	✅	🟡 wire
Short interest	🟡 add-on (Polygon can)	❌ (no product)
Options chains	✅	🟡 thin coverage
Ticker universe & search	✅	🟡 wire

Reading the matrix: Polygon-only has many ❌ in the fundamentals/analyst rows that cannot be resolved. FMP-only has mostly 🟡 (integration work on existing endpoints) and only one ❌ (short interest).

5. Recommendation

- **A single-vendor Polygon build is not achievable** for MarketCatalyst as designed. Polygon should remain the **market-data/price/technical backbone** (and is the *only* one of the two that can add real index/VIX levels and short interest), but it cannot supply estimates, analyst data, ownership or transcripts.
- **A single-vendor FMP build is achievable (~95%+)**. It would consolidate the hard fundamentals we already depend on and let us retire several proxy hacks, at the cost of (a) losing **short interest** entirely, (b) accepting FMP's non-tick real-time and thinner options, and (c) re-implementing the 8-K earnings-announcement logic. If

true real-time and short interest matter, a **Polygon (prices + short interest + indices) + FMP (fundamentals + analyst + ownership)** split — essentially today's model, tightened — remains the most complete option.

- **The lone data point neither vendor solves is short interest** — that needs FINRA (or Polygon's short-interest add-on specifically).

Appendix — source of every claim

This analysis was built from a line-level inventory of the codebase:

- **Polygon:** 17 REST endpoints + 1 WebSocket via `src/vendors/polygon/polygon.service.ts` (Starter plan, 15-min delay, 5-yr history; `NOT_AUTHORIZED` on real-time, indices, short interest, >5yr, stable financials).
- **FMP:** 14 `/stable/` endpoints via `src/vendors/fmp/fmp.service.ts` (v3/v4 deprecated → 403; no short interest; EPS-only surprises; per-firm PT ~30% drop).
- **FRED:** `src/vendors/fred/fred.service.ts` + `src/common/macro-series.ts` (WTI, BTC, 12 macro series, regime inputs; gold = GLD × 10.89, not FRED).
- **SEC-EDGAR:** `src/vendors/sec-edgar/sec-edgar.service.ts` + `src/sync/{sec-form4,sec-13f,edgar-8k,edgar-ipo-pipeline}.job.ts` (Form 4, 13F, 8-K/2.02, IPO pipeline).
- **Data-point catalog:** `MarketCatalystUI/app/iq/types/*.ts` + `app/iq/screens/*.tsx`.

Vendor-catalog capabilities (what each vendor *could* provide beyond what we call today) reflect Polygon.io and FMP published API product lines; the specific FMP `/stable` paths in §3.2 and the Polygon add-ons in §2.2 should be confirmed against current vendor docs and your plan entitlements before commercial commitment.